Report 8

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I. Introduction

In this experiment, we want to design a lead and lag controller for an flexible joint robot.
A flexible joint robot is a type of robot designed with joints that offer some degree of flexibility, compliance, or elasticity, distinguishing them from rigid robots. These robots incorporate elements like springs, elastic actuators, or specially designed mechanisms to allow controlled movement with some give or deformation at the joints.

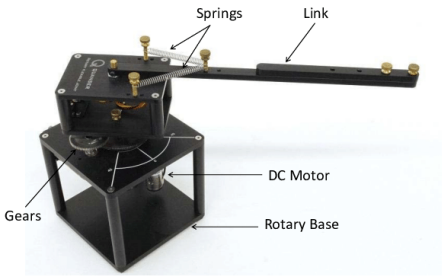


Fig. 1. Flexible Joint Robot

II. Mathematical Modeling

We use the following diagram to derive the mathematical model of the system:

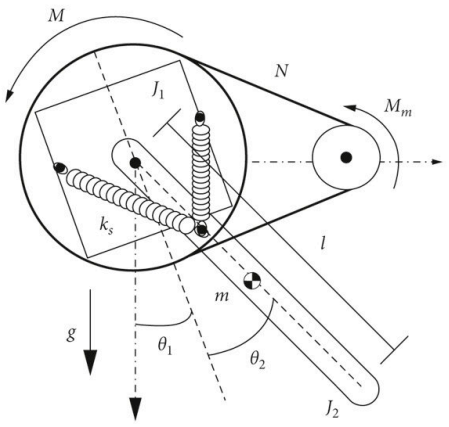


Fig. 2. Flexible Joint Robot Model

Using Lagrange-Hamilton, we arrive at the following equations.

$$J_{eq}\ddot{\theta} + J_{Arm}(\ddot{\theta} + \ddot{\alpha}) = T_{output} - B_{eq}\dot{\theta} \tag{1}$$

$$J_{Arm}(\ddot{\theta} + \ddot{\alpha}) + K_{stiff}\alpha = 0 \tag{2}$$

And we have the torque of a DC motor:

$$T_{output} = \frac{\eta_m \eta_g K_g K_m (V_m - K_g K_m \dot{\theta})}{R_m} \tag{3}$$

The selected state variables are:

$$X = \begin{pmatrix} \theta \\ \dot{\theta} \\ \alpha \\ \dot{\alpha} \end{pmatrix}$$

Then, using the solve command in MATLAB, we solve the above equations and obtain the state-space matrices:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-(K_m K_t n_g n_m K_g^2 + B_{eq} R_m)}{J_{eq} R_m} & \frac{K_{stiff}}{J_{eq}} & 0 \\ 0 & 0 & 0 & 1 \\ g & \frac{(K_m K_t n_g n_m K_g^2 + B_{eq} R_m)}{J_{eq} R_m} & \frac{-K_{stiff}(J_{arm} + J_{eq})}{J_{eq} J_{arm}} & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 \\ \frac{K_t K_g \eta_g \eta_m}{J_{eq} R_m} \\ 0 \\ -\frac{K_t K_g \eta_g \eta_m}{J_{eq} R_m} \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \qquad D = 0$$

III. Desired Poles

The purpose of designing a controller is to satisfy the following conditions:

- Percent overshoot less than 5 percent

$$P.O = 100e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$$

- Rising time less than 250 milliseconds

$$T_r = \frac{\pi - \theta}{\omega_n\sqrt{1-\zeta^2}} \cong \frac{1.8}{\omega_n}$$

Using the equations above, we arrive at the following results:

$$\zeta = 0.6901 \qquad \omega_n = 7.2 \implies P = -4.968 \pm j5.211$$

We also place the two other poles of the system where the conditions of the dominant pole exist.

IV. Using MATLAB to Implement State Feedback

Using the obtained matrices, we design the controller step by step with the desired poles and the following codes.

- place():The given command, using matrices A and B and the desired poles, calculates the coefficients, resulting in a matrix output. The matrix of coefficients obtained is as follows:

$$k = \begin{pmatrix} 3.2941 & 0.1938 & -35.2507 & -1.0548 \end{pmatrix}$$

- $A_{new} = A - Bk$: We multiply the obtained matrix of coefficients by B and subtract it from matrix A.
- New tf:After executing these two steps, we arrive at the new matrix A_{new}, which gives us the new transfer function with the desired poles that we wanted.

V. SIMULINK

We can implement the State Feedback model using Simulink as follows.

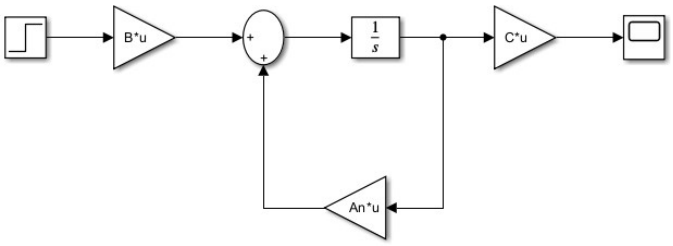


Fig. 3. Simulink

VI. Results

```
      RiseTime: 0.3312
  TransientTime: 0.8879
   SettlingTime: 0.8879
    SettlingMin: 0.4262
    SettlingMax: 0.4931
      Overshoot: 4.4681
     Undershoot: 0
           Peak: 0.4931
       PeakTime: 0.6739
```

Fig. 4. Step info after using controller

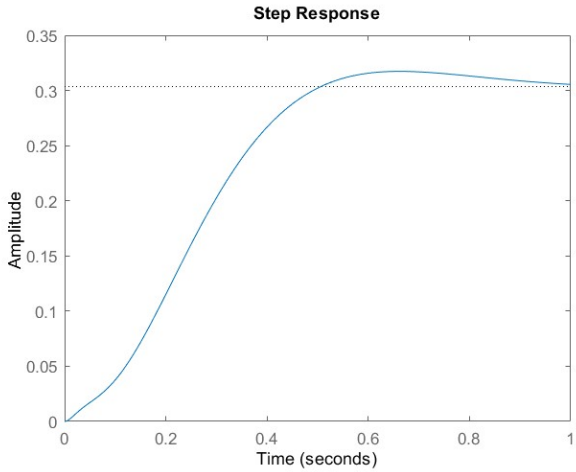


Fig. 5. Step response after using controller

VII. Conclusion

Using state feedback, we can precisely relocate the system's poles, but it has some issues:

- Permanently Disturbing the Steady-State Response: When we manipulate the poles, the steady-state response may be permanently affected.
- Inability to Use State Feedback: Despite its precision, state feedback cannot always be employed due to practical limitations.
- Saturation of the System: If the obtained coefficients exceed a certain limit, the system will saturate.

Based on the results, the overshoot was less than 5 percent, but the rising time was slightly more than the desired amount. This could be because an inaccurate formula was used.